

● HARD

● ALGEBRA

Algebra

10 questions · ~15 min

12

Q1

ALGEBRA

$$\frac{7}{2}x + 6y = 25$$

$$\frac{5}{2}x + 6y = 23$$

The solution to the given system of equations is x, y . What is the value of $\frac{17}{2}x + 18y$?

- A. 2
- B. 3
- C. 48
- D. 71

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x -coordinate of the x -intercept of the graph of $y = hx$ in the xy -plane?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

ALGEBRA

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Q5

ALGEBRA

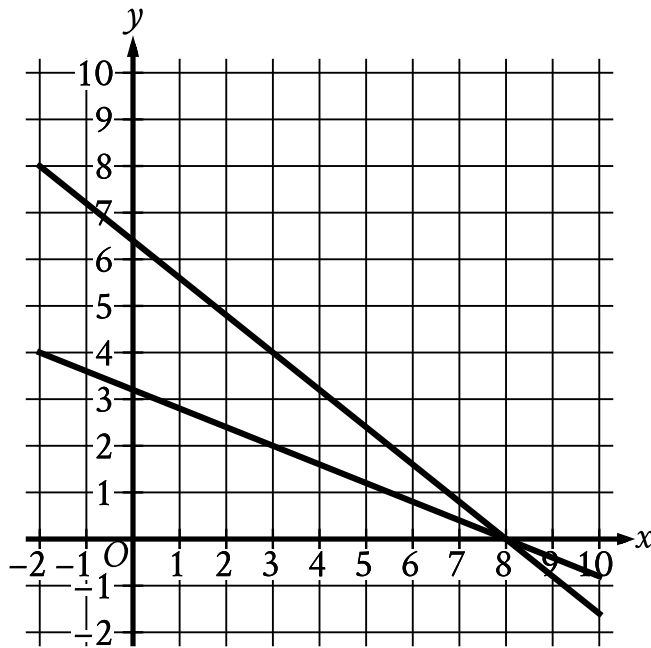
Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

- A. The number of large jars Keenan filled
- B. The number of small jars Keenan filled
- C. The total number of cups of vegetable broth in the large jars
- D. The total number of cups of vegetable broth in the small jars

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A. 3.78
- B. 35.78
- C. 487.89
- D. 519.89



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (3 comma 2)
 - (8 comma 0)
- For the second line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (3 comma 4)
 - (8 comma 0)

What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$
 $-10x - 4y = -64$

B. $8x - 4y = 32$

$$-10x + 4y = -64$$

C. $4x - 10y = 32$

$$-8x + 10y = -64$$

D. $4x + 10y = 32$

$$-8x - 10y = -64$$

Q8

ALGEBRA

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A. $10x + 7y = 1$

$$ax - 2by = 1$$

B. $10x + 7y = 1$

$$ax + 2by = 1$$

C. $10x + 7y = 1$

$$2ax + by = 1$$

D. $5x - 7y = 1$

$$ax + by = 1$$

$$\begin{aligned}7x - 5y &= 4 \\4x - 8y &= 9\end{aligned}$$

If (x, y) is the solution to the system of equations above, what is the value of $3x + 3y$?

- A. -13
- B. -5
- C. 5
- D. 13

x	y
3	7
k	11
12	n

The table above shows the coordinates of three points on a line in the xy -plane, where k and n are constants. If the slope of the line is 2, what is the value of $k+n$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.